MED 520 Examination

General Pathology and Disease Mechanisms – Answer Key

1. (B) Hypertrophy.
Hypertrophy is an increase in cell size, often in response to increased workload. Example: cardiac muscle hypertrophy in hypertension. In contrast, hyperplasia is an increase in cell number.

2. (C) Caseous necrosis.
Caseous necrosis has a distinctive cheese-like appearance and is characteristic of granulomatous inflammation, particularly in tuberculosis. It combines features of coagulative and liquefactive necrosis.

3. (C) Neutrophils.
Neutrophils (polymorphonuclear leukocytes) are the first responders in acute inflammation, arriving within minutes to hours. They phagocytose bacteria and debris. Macrophages arrive later in acute inflammation and dominate chronic inflammation.

4. (C) AFP (Alpha-Fetoprotein).
AFP is elevated in hepatocellular carcinoma and certain germ cell tumors. PSA is specific for prostate cancer, CA-125 for ovarian cancer, and CEA is associated with colorectal and other cancers.

5. *Sample answer:*
Benign tumors:

- **Growth**: Slow, expansile, often encapsulated
- **Invasion**: Do not invade surrounding tissues
- **Metastasis**: Do not metastasize
- **Differentiation**: Well-differentiated, resemble tissue of origin
- Example: Lipoma (benign adipose tumor)

Malignant tumors:

- **Growth**: Rapid, infiltrative
- **Invasion**: Invade adjacent tissues and destroy normal architecture
- **Metastasis**: Spread to distant sites via lymphatics and blood
- **Differentiation**: Poorly differentiated (anaplastic) to well-differentiated
- Example: Liposarcoma (malignant adipose tumor)

Why metastasis is critical: Metastasis is the primary cause of cancer-related mortality. Local tumors can often be surgically removed, but metastatic disease is difficult to treat and leads to multi-organ failure. The ability to metastasize definitively classifies a tumor as malignant.

6. *Sample answer:*
Pathophysiology: Atherosclerosis is characterized by the formation of atherosclerotic plaques (atheromas) in arterial walls, leading to stenosis, thrombosis, and ischemia.

Major risk factors:

- Hyperlipidemia (especially LDL cholesterol)
- Hypertension
- Smoking
- Diabetes mellitus
- Age, male sex, family history

Mechanisms:

(a) **Endothelial dysfunction**: Risk factors damage endothelium, increasing permeability and promoting inflammation

(b) **Lipid accumulation**: LDL cholesterol infiltrates the intima and becomes oxidized (ox-LDL)

(c) **Inflammation**: Monocytes migrate into the intima, differentiate into macrophages, and ingest ox-LDL, becoming foam cells

(d) **Smooth muscle proliferation**: Growth factors stimulate smooth muscle cell migration and proliferation, forming a fibrous cap

(e) **Plaque formation**: Lipid core (foam cells, extracellular lipid) covered by fibrous cap. Plaques can rupture, causing thrombosis

Complications include myocardial infarction, stroke, and peripheral arterial disease.

7. *Sample answer:*
Apoptosis: Programmed cell death characterized by controlled, energy-dependent dismantling of the cell without inflammation.

Differences from necrosis:

- **Apoptosis**: Cell shrinkage, chromatin condensation, membrane blebbing, formation of apoptotic bodies, phagocytosis without inflammation
- **Necrosis**: Cell swelling, membrane rupture, release of contents, inflammatory response

Two major pathways:

(a) **Intrinsic (mitochondrial) pathway**: Triggered by DNA damage, oxidative stress, or growth factor withdrawal. Mitochondria release cytochrome c, activating caspases

(b) **Extrinsic (death receptor) pathway**: Triggered by binding of ligands (e.g., Fas ligand, TNF) to death receptors, directly activating caspases

Role of caspases: Caspases are proteases that execute apoptosis by cleaving cellular proteins, leading to DNA fragmentation, cytoskeletal breakdown, and cell dismantling.

Significance:

- **Development**: Eliminates unwanted cells (e.g., webbing between fingers)
- **Homeostasis**: Balances cell proliferation
- **Disease**: Excessive apoptosis causes atrophy (e.g., neurodegenerative diseases); deficient apoptosis contributes to cancer

8. *Sample answer:*
Carcinogenesis: Multi-step process where normal cells progressively acquire genetic and epigenetic alterations, leading to malignancy.

Multi-step model:

(a) **Initiation**: DNA damage by carcinogens (chemicals, radiation, viruses) creates mutations

(b) **Promotion**: Clonal expansion of initiated cells; mutations accumulate

(c) **Progression**: Additional mutations lead to invasion and metastasis

Key genetic changes:

- **Oncogenes**: Mutated proto-oncogenes (e.g., RAS, MYC) promote uncontrolled proliferation. Gain-of-function mutations; dominant (one mutated allele sufficient)
- **Tumor suppressor genes**: (e.g., TP53, RB) normally inhibit growth or promote apoptosis. Loss-of-function mutations; recessive (both alleles must be inactivated, "two-hit hypothesis")
- **Apoptosis regulators**: (e.g., BCL-2 family) defects prevent elimination of damaged cells
- **DNA repair genes**: (e.g., BRCA1, BRCA2, mismatch repair genes) mutations increase mutation rate, accelerating carcinogenesis

Example: Colorectal cancer progression: normal epithelium → adenoma (APC mutation) → carcinoma (KRAS, TP53 mutations) → metastasis.

Cancer results from the accumulation of multiple mutations affecting these pathways, conferring hallmarks like sustained proliferation, evasion of apoptosis, angiogenesis, and metastatic capability.